NATIONAL CURRICULUM DEVELOPMENT CENTRE (NCDC) UGANDA

LOWER SECONDARY COMPETENCY-BASED CURRICULUM (CBC)

CONSOLIDATED CURRICULUM STRUCTURE (O-LEVEL: SENIOR 1 – SENIOR 4)

Subjects: Mathematics, Physics, Chemistry, Biology

1. SENIOR 1 CURRICULUM SYLLABUS

1.1 MATHEMATICS (SENIOR 1)

- **Theme:** Number & Algebra / Geometry & Measures / Data Handling
- **Key Learning Outcomes & Competencies:**
    - **Number Bases:** Identify numbers in different bases using manipulative tools (abacus, place value charts); convert numbers from base ten to other bases (base 2, 5, 8) and vice versa; apply binary numbers in digital contexts.
    - **Working with Integers:** Perform basic operations (\(+,-,\times ,\div\)) on directed numbers; apply order of operations (BODMAS) to real-life financial and temperature scenarios.
    - **Fractions, Decimals & Percentages:** Express fractions as decimals and percentages; solve practical problems involving direct/indirect proportions and percentage profit/loss.
    - **Rectangular Cartesian Coordinates:** Plot points in all four quadrants; draw and interpret geometric shapes on coordinate planes.
    - **Geometric Construction:** Construct perpendicular bisectors, angle bisectors, angles (\(6{0}^{\circ },9{0}^{\circ },4{5}^{\circ },3{0}^{\circ }\)), and triangles using a ruler and pair of compasses.
    - **Sequences and Patterns:** Identify number patterns, formulate rule expressions for nth terms, and apply sequences in real-life arrangements.
    - **Bearings:** Determine directions using cardinal points and 3-figure bearings; apply scale drawing to solve distance and direction problems.

- **General Angle Properties:** Apply angle properties of straight lines, parallel lines, intersecting lines, and polygons.
- **Data Collection and Presentation:** Collect raw primary/secondary data; organize data into tally tables and frequency distribution tables; present data using pictograms, bar charts, and pie charts.
- **Reflection:** Determine images of figures under line reflection; state properties of reflection in coordinate geometry.
- **Equation of Lines and Linear Equations:** Solve simple linear equations in one unknown; represent linear equations on a Cartesian plane.
- **Algebra 1:** Basic algebraic expressions, expansion, factorization of linear terms, and substitution.
- **Business Arithmetic:** Calculate simple interest, profit and loss, discount, commission, and currency conversions relevant to local Ugandan markets.
- **Time and Time Tables:** Read 12-hour and 24-hour clocks; calculate elapsed time and interpret travel/bus timetables.

1.2 PHYSICS (SENIOR 1)

- **Theme:** General Physics / Thermal Physics / Light
- **Key Learning Outcomes & Competencies:**
  - **Introduction to Physics:** Define physics, identify its branches, and recognize its application in everyday life and Ugandan technology/industry; observe laboratory safety rules.
  - **Measurements in Physics:** Measure fundamental quantities (length, mass, time) and derived quantities (area, volume, density) using appropriate SI instruments (vernier calipers, micrometers, measuring cylinders).
  - **States of Matter:** Explain the kinetic theory of matter; demonstrate diffusion, Brownian motion, and phase changes (solid, liquid, gas).
  - **Effects of Forces:** Identify contact and non-contact forces; demonstrate weight, friction, tension, upthrust, and surface tension; apply Hooke's Law (\(F=kx\)).
  - **Temperature Measurement:** Distinguish between heat and temperature; construct liquid-in-glass thermometers and calibrate them on Celsius and Kelvin scales.
  - **Heat Transfer:** Investigate conduction, convection, and radiation in domestic applications (charcoal stoves, thermoflasks, solar heaters).
  - **Expansion of Solids, Liquids, and Gases:** Demonstrate thermal expansion; analyze bimetallic strips and the anomalous expansion of water and its ecological importance.
  - **Nature of Light and Rectilinear Propagation:** Demonstrate that light travels in straight lines; construct pinhole cameras; explain shadows and solar/lunar eclipses.

1.3 CHEMISTRY (SENIOR 1)

- **Theme:** Chemistry and Society / Matter / Air and Water / Earth Materials
- **Key Learning Outcomes & Competencies:**

- **Chemistry and Society:** Define chemistry; identify chemical products used in daily life (soaps, medicine, food, fuels); discuss career opportunities and safety in the laboratory.
- **Experimental Chemistry:** Select, set up, and safely use common laboratory apparatus; carry out basic measurement, separation, and heating techniques.
- **States and Changes of Matter:** Describe physical and chemical changes; investigate melting, boiling, evaporation, condensation, freezing, and sublimation using local examples.
- **Using Materials:** Classify materials based on physical properties (metals, non-metals, ceramics, polymers, composites) and justify their selection for local construction and domestic uses.
- **Temporary and Permanent Changes:** Distinguish reversible (physical) from irreversible (chemical) changes through practical experimentation.
- **Elements, Compounds, and Mixtures:** Classify matter into elements, compounds, and mixtures; separate mixtures using filtration, evaporation, simple/fractional distillation, chromatography, and crystallization.
- **Air:** Determine the percentage composition of oxygen in air; demonstrate the preparation, test, and properties of oxygen; investigate air pollution and combustion.
- **Water:** Test for purity of water; explain the water cycle, water treatment processes, and methods for purifying contaminated water for community usage.
- **Rocks and Minerals:** Classify igneous, sedimentary, and metamorphic rocks; investigate local mineral resources in Uganda and their economic value.

1.4 BIOLOGY (SENIOR 1)

- **Theme:** Diversity of Living Things / Cell Biology / Classification
- **Key Learning Outcomes & Competencies:**
  - **Introduction to Biology:** Define biology, state its branches, and appreciate its relevance to medicine, agriculture, forestry, and environmental conservation in Uganda.
  - **Cells:** Use light microscopes to observe plant and animal cells; draw and label cell structures; outline functions of cell organelles; differentiate plant cells from animal cells.
  - **Classification:** Explain the necessity of classifying organisms; use dichotomous keys to identify organisms based on observable external features.
  - **The Five Kingdoms of Life:** Classify living organisms into Monera, Protista, Fungi, Plantae, and Animalia; identify key characteristic features of each kingdom.
  - **Viruses:** Explain the structure of viruses; discuss viral transmission, impact on human/plant health (HIV, Ebola, COVID-19, cassava mosaic), and preventive measures.
  - **Insects:** Examine external structures of economic/medical insects (mosquito, housefly, honeybee, cockroach, grasshopper); discuss their role as vectors or pollinators and life cycles (complete vs. incomplete metamorphosis).
  - **Flowering Plants:** Examine structures of monocotyledonous and dicotyledonous roots, stems, leaves, and flowers; differentiate monocots from dicots.

2. SENIOR 2 CURRICULUM SYLLABUS

2.1 MATHEMATICS (SENIOR 2)

- **Theme:** Relations & Functions / Geometry / Algebra / Trigonometry
- **Key Learning Outcomes & Competencies:**
    - **Mappings and Relations:** Represent relations using arrow diagrams, tables, graph plots, and ordered pairs; identify domain, co-domain, range, and types of mappings (one-to-one, many-to-one).
    - **Vectors and Translation:** Define vectors; calculate magnitude and direction of 2D vectors; perform vector addition/subtraction; apply translation vectors in coordinate planes.
    - **Graphs:** Plot linear and quadratic graphs; read gradients, intercepts, and solve equations graphically.
    - **Indices and Standard Form:** Apply laws of indices (\({a}^{m}\times {a}^{n}={a}^{m+n}\), \(({a}^{m}{)}^{n}={a}^{mn}\), etc.); write numbers in standard form (\(A\times 1{0}^{n}\)).
    - **Inequalities and Regions:** Represent linear inequalities on number lines and graph Cartesian planes; determine shaded region solutions.
    - **Algebra 2 (Quadratic Expressions):** Expand binomial expressions; factorize quadratic expressions of the form \(a{x}^{2}+bx+c\).
    - **Similarities and Enlargement:** Determine scale factors of enlargement; calculate areas and volumes of similar figures using area and volume scale factors.
    - **Circles (Circumference and Area):** Calculate arc length, perimeter of sectors, sector areas, and segment areas.
    - **Rotation:** Rotate 2D figures about a center of rotation through given angles; determine center and angle of rotation.
    - **Length and Area (Polygons):** Calculate areas of triangles, quadrilaterals, parallelograms, trapeziums, and regular polygons.
    - **Nets, Areas, and Volumes of Solids:** Construct nets of 3D solids (prisms, pyramids, cylinders, cones, spheres); calculate surface area and volume of solids.
    - **Surds:** Simplify surds (\(a\sqrt{b}\)); perform addition, subtraction, multiplication, and rationalization of denominators.
    - **Set Theory:** Use Venn diagrams to solve 2-set and 3-set problems; determine union, intersection, and complement of sets.

2.2 PHYSICS (SENIOR 2)

- **Theme:** Mechanics / Optics / Electricity & Magnetism / Space Science
- **Key Learning Outcomes & Competencies:**
    - **Work, Energy, and Power:** Define work (\(W=F\cdot d\)), mechanical energy (K.E. \(=\frac{1}{2}m{v}^{2}\), P.E. \(=mgh\)), and power (\(P=\frac{W}{t}\)); demonstrate law of conservation of energy; perform calculations.
    - **Turning Effect of Forces (Moments):** Apply the principle of moments (\(\sum \text{Clockwise} = \sum \text{Anticlockwise}\)); locate center of gravity; determine conditions for stability of structures.

- **Pressure in Solids and Fluids:** Calculate pressure ($P=\frac{F}{A}$ and $P=h\rho g$); explain atmospheric pressure, barometers, hydraulic systems, and Archimedes' principle.
- **Mechanical Properties of Materials:** Investigate tensile strength, elasticity, plasticity, ductility, and stiffness in everyday engineering materials.
- **Reflection of Light at Curved Surfaces:** Construct ray diagrams for concave and convex mirrors; calculate focal length, image distance, and magnification ($m=\frac{v}{u}$); state practical applications (car headlights, shaving mirrors).
- **Magnets and Magnetism:** Map magnetic field patterns around bar magnets; explain domains, magnetization/demagnetization, and terrestrial magnetism.
- **Electrostatics:** Demonstrate electrocharging by friction, induction, and contact; operate gold-leaf electroscopes; explain lightning and lightning conductors.
- **The Solar System:** Describe the planetary system, gravitational force, motion of planets, satellites, and space exploration.

2.3 CHEMISTRY (SENIOR 2)

- **Theme:** Chemical Reactions / Periodic Table / Quantitative Chemistry
- **Key Learning Outcomes & Competencies:**
  - **Acids and Alkalis:** Define acids/alkalis using local examples; test pH using indicators (litmus, universal indicator, plant-extracted indicators); carry out neutralization reactions and applications.
  - **Salts:** Prepare soluble and insoluble salts using precipitation, neutralization, and acid-metal/carbonate reactions; write balanced chemical equations with state symbols.
  - **The Periodic Table:** Explain the arrangement of elements in Groups and Periods based on atomic number and electronic configurations; observe trends in Group I (alkali metals), Group VII (halogens), and Group VIII (noble gases).
  - **Carbon in the Environment:** Examine carbon cycles; describe allotropes of carbon (diamond, graphite, amorphous carbon); prepare and test carbon dioxide gas ($C{O}_{2}$) and examine greenhouse effects.
  - **The Reactivity Series:** Investigate reactions of metals (K, Na, Ca, Mg, Al, Zn, Fe, Cu) with water, steam, oxygen, and dilute acids; arrange metals in order of reactivity.

2.4 BIOLOGY (SENIOR 2)

- **Theme:** Soil / Plant & Animal Physiology / Transport
- **Key Learning Outcomes & Competencies:**
  - **Physical and Chemical Properties of Soil:** Analyze soil texture, structure, soil air, moisture content, organic matter content, soil pH, porosity, and water retention capacity through field investigations.
  - **Soil Erosion and Conservation:** Identify agents and types of soil erosion; execute sustainable soil management/conservation practices (terracing, mulching, contour farming).
  - **Nutrition Types and Nutrient Compounds:** Differentiate autotrophic from heterotrophic nutrition; perform food tests for starch, reducing/non-reducing sugars, proteins, lipids, and Vitamin C.

- **Nutrition in Green Plants:** Investigate requirements for photosynthesis (light, chlorophyll, $C{O}_{2}$, water); examine internal leaf structure adaptations; test a leaf for starch.
- **Nutrition in Mammals:** Describe human dentition, digestive tract anatomy, enzymatic digestion of carbohydrates, proteins, and fats, absorption in the ileum, and assimilation.
- **Transport in Plants:** Explain the mechanisms of water uptake and mineral salt absorption; investigate xylem transport, transpiration rate using potometers, and phloem translocation.
- **Transport in Animals:** Describe the double circulatory system of humans; detail the structure and function of the heart, blood vessels (arteries, veins, capillaries), blood cells, and immune response.

3. SENIOR 3 CURRICULUM SYLLABUS

3.1 MATHEMATICS (SENIOR 3)

- **Theme:** Coordinate Geometry / Trigonometry / Matrices / Probability
- **Key Learning Outcomes & Competencies:**
  - **Equation of a Straight Line:** Determine gradient ($m=\frac{{y}_{2}-{y}_{1}}{{x}_{2}-{x}_{1}}$), y-intercept, parallel/perpendicular line equations, and midpoint of line segments.
  - **Trigonometry 1:** Apply sine, cosine, and tangent ratios ($\sin ,\cos ,\tan$) in right-angled triangles to solve angle of elevation and depression problems.
  - **Data Collection and Display (Advanced):** Construct grouped frequency tables; estimate mean, median, and modal class for grouped data; plot histograms and cumulative frequency curves (ogives).
  - **Vectors (Advanced):** Apply vector geometry to solve geometric proofs in figures; calculate position vectors and ratio division of lines.
  - **Ratios and Proportions:** Solve problems involving direct, inverse, and compound proportions, scale factors, and unit rates.
  - **Business Mathematics:** Calculate compound interest, appreciation, depreciation, income tax bands, PAYE, and utility bill calculations.
  - **Trigonometry 2:** Apply the Sine Rule ($\frac{a}{\sin A}=\frac{b}{\sin B}$) and Cosine Rule ($\({a}^{2}={b}^{2}+{c}^{2}-2bc\cos A$) to non-right-angled triangles.
  - **Matrices:** Perform matrix operations (addition, subtraction, scalar multiplication, matrix multiplication); calculate determinants and inverses of 2×2 matrices.
  - **Matrix Transformations:** Represent 2D transformations (reflection, rotation, enlargement, shear, stretch) using 2×2 transformation matrices.
  - **Simultaneous Equations:** Solve systems of two linear equations algebraically (substitution, elimination) and graphically.
  - **Probability:** Calculate experimental and theoretical probability; construct tree diagrams and Venn diagrams for combined/independent/mutually exclusive events.
  - **Quadratic Equations:** Solve quadratic equations by factorization, completing the square, and using the quadratic formula ($x=\frac{-b\pm \sqrt{{b}^{2}-4ac}}{2a}$).
  - **Circle Properties:** State and prove circle theorems involving subtended angles, cyclic quadrilaterals, tangents, and alternate segments.

3.2 PHYSICS (SENIOR 3)

- **Theme:** Dynamics & Kinematics / Waves & Sound / Astrophysics
- **Key Learning Outcomes & Competencies:**
    - **Linear and Non-Linear Motion:** Graph distance-time and velocity-time curves; derive and apply equations of uniformly accelerated motion (\(v=u+at\), \(s=ut+\frac{1}{2}a{t}^{2}\), \({v}^{2}={u}^{2}+2as\)); calculate momentum (\(p=mv\)) and apply Newton's Laws of Motion.
    - **Refraction, Dispersion, and Total Internal Reflection:** Determine refractive index (\(n=\frac{\sin i}{\sin r}\)); demonstrate critical angle and total internal reflection in optical fibers and prisms; explain dispersion of white light.
    - **Lenses and Optical Instruments:** Construct ray diagrams for convex and concave lenses; solve thin lens formula (\(\frac{1}{f}=\frac{1}{u}+\frac{1}{v}\)); describe astronomical telescopes, microscopes, and human eye corrections.
    - **General Wave Properties:** Differentiate transverse from longitudinal waves; calculate wave velocity (\(v=f\lambda\)); demonstrate wave reflection, refraction, diffraction, and interference using ripple tanks.
    - **Sound Waves:** Investigate production, transmission, speed of sound in media, echoes, reverberation, and applications of ultrasound in medicine and sonar navigation.
    - **Heat Quantities and Specific Heat Capacity:** Measure heat capacities using calorimetry; apply \(Q=mc\Delta T\) and \(Q=mL\); distinguish specific heat capacity from latent heat of fusion/vaporization.
    - **Stars and Galaxies:** Describe stellar evolution, main sequence stars, supernovae, black holes, and the structure of the Milky Way and expanding universe.
    - **Satellites and Communication:** Differentiate natural from artificial satellites; explain geostationary orbits, GPS systems, and telecommunication infrastructure in Uganda.

3.3 CHEMISTRY (SENIOR 3)

- **Theme:** Organic Chemistry / Bonding / Stoichiometry / Chemical Kinetics
- **Key Learning Outcomes & Competencies:**
    - **Carbon in Life (Organic Chemistry):** Structure and nomenclature of alkanes, alkenes, alkynes, and alcohols; fractional distillation of crude oil; preparation of biogas and ethanol via fermentation.
    - **Structures and Bonds:** Explain ionic, covalent, metallic, and dative bonding; correlate giant ionic, simple molecular, giant covalent, and metallic structures to melting points and conductivity.
    - **Formulae, Stoichiometry, and the Mole Concept:** Define the mole, Avogadro's constant, molar mass, and empirical/molecular formulae; calculate concentration (\(mol/d{m}^{3}\)), reacting masses, and gaseous volumes at STP/RTP.
    - **Properties and Structure of Matter:** Investigate physical properties of giant structures (diamond, graphite, silicon dioxide, sodium chloride) based on bonding forces.
    - **Fossil Fuels:** Evaluate environmental impacts of burning fossil fuels; propose sustainable alternative biofuel solutions suitable for Ugandan rural/urban areas.

- **Chemical Reactions (Rates and Energy):** Determine factors affecting reaction rates (concentration, temperature, surface area, catalysts); construct energy level diagrams for exothermic and endothermic reactions (\(\Delta H\)).

3.4 BIOLOGY (SENIOR 3)

- **Theme:** Respiration / Gas Exchange / Homeostasis / Coordination
- **Key Learning Outcomes & Competencies:**
  - **Gaseous Exchange:** Compare structures of gas exchange surfaces in ameba, insects (tracheoles), fish (gills), amphibians (skin/mouth/lungs), and mammals (alveoli); demonstrate mechanics of breathing.
  - **Respiration:** Compare aerobic (\({C}_{6}{H}_{12}{O}_{6}+6{O}_{2}\rightarrow 6C{O}_{2}+6{H}_{2}O+38ATP\)) and anaerobic respiration (lactic acid vs. alcoholic fermentation); construct simple respirometers.
  - **Excretion and Homeostasis:** Explain excretion in lower organisms and humans; detail kidney nephron structure, ultrafiltration, selective reabsorption, osmoregulation, and skin temperature control.
  - **Coordination in Plants and Animals:** Investigate plant tropisms (phototropism, geotropism); examine human nervous control (neurons, brain, spinal cord, reflex arc) and endocrine/hormonal regulation.
  - **Support and Movement:** Differentiate exoskeletons, endoskeletons, and hydrostatic skeletons; examine locomotion mechanisms, bones of the axial/appendicular skeleton, and skeletal muscle action.

4. SENIOR 4 CURRICULUM SYLLABUS

4.1 MATHEMATICS (SENIOR 4)

- **Theme:** Advanced Functions / Linear Programming / 3D Geometry
- **Key Learning Outcomes & Competencies:**
  - **Composite and Inverse Functions:** Evaluate composite functions \(f(g(x))\) or \(g(f(x))\); derive inverse functions \({f}^{-1}(x)\) algebraically and graphically.
  - **Equations and Inequalities (Advanced):** Solve systems involving linear and non-linear simultaneous equations; apply sign charts to solve quadratic inequalities.
  - **Linear Programming:** Formulate linear inequalities from real-world constraint word problems; plot feasible regions; determine optimal solutions using objective functions/search lines.
  - **Loci:** Construct locus of points equidistant from two fixed points, a fixed line, or a single point; solve locus intersection problems.
  - **Lines and Planes in Three Dimensions (3D Geometry):** Calculate angles between two planes, angles between a line and a plane, and lengths of space diagonals in cuboids, pyramids, and prisms.

4.2 PHYSICS (SENIOR 4)

- **Theme:** Current Electricity / Electromagnetism / Modern Physics
- **Key Learning Outcomes & Competencies:**
    - **Introduction to Current Electricity:** Define electric current, charge ($Q=It$), electromotive force (e.m.f.), potential difference (V), and resistance; set up series and parallel circuits.
    - **Voltage, Resistance, and Ohm's Law:** State Ohm's Law ($V=IR$); determine internal resistance of cells; calculate equivalent resistance in complex circuit networks.
    - **Electromagnetic Effects:** Demonstrate magnetic fields produced by electric currents in straight wires, solenoids, and coils; explain operation of electric motors, d.c. generators, step-up/step-down transformers, and Fleming's left/right-hand rules.
    - **Electric Energy Distribution and Safety:** Calculate electrical power ($P=VI={I}^{2}R$) and energy consumption (kWh); explain grid distribution, fuses, earthing, circuit breakers, and domestic wiring.
    - **Atomic Models:** Trace historical atomic models (Rutherford, Bohr); describe atomic structure, isotopes, atomic number (Z), and mass number (A).
    - **Nuclear Processes:** Differentiate nuclear fission from fusion; write balanced nuclear equations for alpha (α), beta (β), and gamma (γ) radiation decay; calculate half-life (${T}_{1/2}$).
    - **Digital Electronics:** Demonstrate basic logic gates (AND, OR, NOT, NAND, NOR) using truth tables; design simple electronic control circuits for automated systems.

4.3 CHEMISTRY (SENIOR 4)

- **Theme:** Electrochemistry / Industrial Chemistry / Chemical Analysis
- **Key Learning Outcomes & Competencies:**
    - **Oxidation and Reduction Reactions (Redox):** Define oxidation/reduction in terms of oxygen transfer, hydrogen transfer, electron transfer, and change in oxidation state; write half-equations for electrolysis processes.
    - **Industrial Processes:** Detail large-scale industrial manufacture of sulfuric acid (Contact Process), ammonia (Haber Process), nitric acid, and extraction of reactive metals (aluminum via electrolysis of bauxite; iron in blast furnace).
    - **Trends in the Periodic Table (Advanced):** Predict chemical behavior across Period 3 and down Groups based on atomic radii, ionization energy, electronegativity, and shield effects.
    - **Energy Changes During Reactions:** Perform calorimetry experiments to calculate enthalpy change of combustion, neutralization, and solution ($\Delta H=-mc\Delta T/n$).
    - **Chemicals for Consumer Products:** Detail the saponification process in soap making; formulate synthetic detergents; analyze chemical composition of cosmetics, fertilizers, and pharmaceuticals.
    - **Nuclear Processes in Chemistry:** Differentiate chemical reactions from nuclear reactions; explain radioisotope applications in carbon dating, medical diagnostics, and industrial tracing.

4.4 BIOLOGY (SENIOR 4)

- **Theme:** Reproduction / Genetics & Evolution / Ecology
- **Key Learning Outcomes & Competencies:**

- **Asexual Reproduction in Plants:** Investigate natural vegetative propagation (tubers, rhizomes, corms, bulbs, runners) and artificial methods (cutting, grafting, layering, marcotting) for agricultural enhancement.
- **Sexual Reproduction in Plants:** Describe flower structure, micro/megasporogenesis, pollination mechanisms, double fertilization, seed formation, and fruit dispersal adaptations.
- **Sexual Reproduction in Humans:** Examine male and female reproductive systems, gametogenesis, menstrual cycle hormonal regulation, fertilization, implantation, pregnancy, fetal development, and sexually transmitted infections (STIs/HIV).
- **Inheritance:** Explain Mendelian genetics (monohybrid crosses, dominant/recessive alleles, genotypes/phenotypes); construct genetic diagrams and Punnett squares; explain sex determination and sex-linked traits (hemophilia, color blindness).
- **Variation and Selection:** Differentiate continuous from discontinuous variation; evaluate mechanisms of natural selection, artificial selection, mutation, and evolutionary adaptations.
- **Concept of Ecology:** Define ecological terms (ecosystem, habitat, niche, population, community); measure abiotic factors and estimate population sizes using quadrats, transects, and capture-recapture methods.
- **Food Chains and Food Webs:** Construct energy pyramids, biomass pyramids, and food webs; trace bioaccumulation of toxic materials in food chains.
- **Associations in Biology:** Investigate symbiotic relationships including mutualism, commensalism, parasitism, and saprophytism with local Ugandan ecological examples.
- **Humans and the Natural Environment:** Assess human impacts on ecosystems (deforestation, wetland degradation, pollution); plan sustainable conservation and waste management projects.

**SASA Platform User Stories **

This shall be used to generate functional requirements

As a teacher, I want to …….. In the app

What do you want to see, what do you want to do, what will make you say the app is good.. What apps have you tried, what gaps do they have….